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CANDIDATE NUMBER

SYDNEY GRAMMAR SCHOOL



2019 Trial Examination

FORM VI

MATHEMATICS 2 UNIT

Monday 12th August 2019

General Instructions

- Reading time — 5 minutes
- Writing time — 3 hours
- Write using black pen.
- NESA-approved calculators and templates may be used.

Total — 100 Marks

- All questions may be attempted.

Section I — 10 Marks

- Questions 1–10 are of equal value.
- Record your answers to the multiple choice on the sheet provided.

Section II — 90 Marks

- Questions 11–16 are of equal value.
- All necessary working should be shown.
- Start each question in a new booklet.

Collection

- Write your candidate number on each answer booklet and on your multiple choice answer sheet.
- Hand in the booklets in a single well-ordered pile.
- Hand in a booklet for each question in Section II, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Write your candidate number on this question paper and hand it in with your answers.
- Place everything inside the answer booklet for Question Eleven.

Checklist

- SGS booklets — 6 per boy
- Multiple choice answer sheet
- Reference Sheet
- Candidature — 90 boys

Examiner

PKH

SECTION I - Multiple Choice

Answers for this section should be recorded on the separate answer sheet handed out with this examination paper.

QUESTION ONE

What is the value of $\log_e 5 + \log_e 50$, correct to two decimal places?

- (A) 5.52
- (B) 2.08
- (C) 4.01
- (D) 2.40

QUESTION TWO

What is the derivative of $y = e^{2x+1}$?

- (A) $y' = (2x + 1)e^{2x+2}$
- (B) $y' = e^{2x+1}$
- (C) $y' = 2e^{2x+1}$
- (D) $y' = \frac{1}{2}e^{2x+1}$

QUESTION THREE

What is the equation of the line parallel to the y -axis passing through the point $(-3, 5)$?

- (A) $x = -3$
- (B) $x = 5$
- (C) $y = -3$
- (D) $y = 5$

QUESTION FOUR

What is the natural domain of the function $y = \frac{1}{\sqrt{2x+1}}$?

- (A) $x > -\frac{1}{2}$
- (B) $x < -\frac{1}{2}$
- (C) $x \leq -\frac{1}{2}$
- (D) $x \geq -\frac{1}{2}$

QUESTION FIVE

Which of the following is equivalent to $\frac{1}{2\sqrt{5}+3}$?

(A) $\frac{2\sqrt{5}+3}{-11}$

(B) $\frac{2\sqrt{5}-3}{17}$

(C) $\frac{2\sqrt{5}-3}{11}$

(D) $\frac{2\sqrt{5}+3}{11}$

QUESTION SIX

Which expression is equivalent to $3^x \times 2^x + 3x^{-2}$?

(A) $6^{2x} + \frac{1}{3x^2}$

(B) $6^x + \frac{1}{9x^2}$

(C) $6^{2x} + \frac{3}{x^2}$

(D) $6^x + \frac{3}{x^2}$

QUESTION SEVEN

What are the correct solutions to the equation $e^{10x} - 11e^{5x} + 10 = 0$?

(A) $x = 1$ and $x = 10$

(B) $x = 1$ and $x = \log_e 10$

(C) $x = 0$ and $x = \log_e 10$

(D) $x = 0$ and $x = \frac{1}{5} \log_e 10$

QUESTION EIGHT

How many solutions does the equation $6x \left(\frac{1}{x} + 1 \right) = 0$ have?

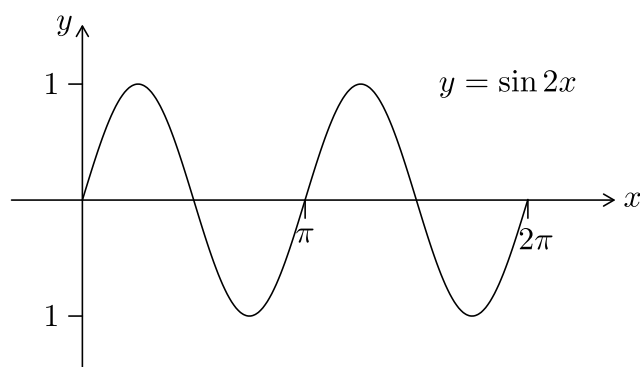
- (A) 0
- (B) 1
- (C) 2
- (D) 3

QUESTION NINE

Which is a correct expression for $-x^2 - 6x - 5$?

- (A) $4 + (x + 3)^2$
- (B) $(x - 3)^2 - 4$
- (C) $4 - (x + 3)^2$
- (D) $-4 + (x + 3)^2$

QUESTION TEN



How many solutions does the equation $5 \sin 2x = x$ for $0 \leq x \leq 2\pi$ have?

- (A) 2
- (B) 3
- (C) 4
- (D) 5

_____ End of Section I _____

Examination continues next page ...

SECTION II - Written Response

Answers for this section should be recorded in the booklets provided.

Show all necessary working.

Start a new booklet for each question.

QUESTION ELEVEN (15 marks) Use a separate writing booklet.

Marks

(a) Solve the inequation $2 - 3x \leq 11$.

1

(b) For the arithmetic sequence $12, 6, 0, \dots$ find the 41st term.

2

(c) Differentiate:

(i) $y = 3x - \frac{1}{x}$

1

(ii) $y = \tan(2x + 1)$

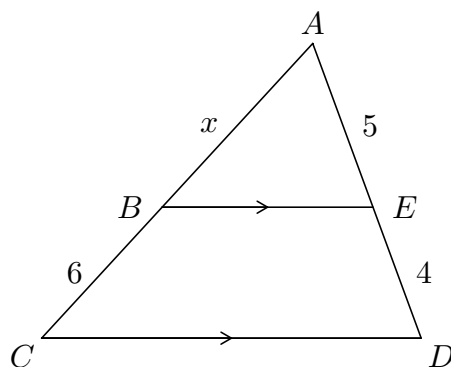
1

(iii) $y = \ln(x^2 + 1)$

1

(d)

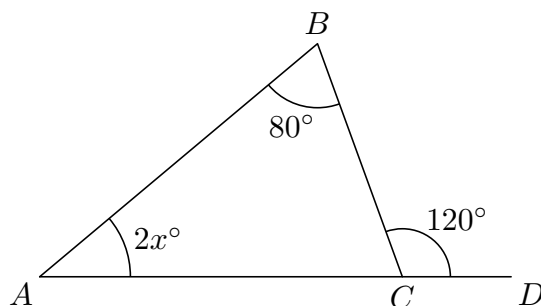
1



Find the value of x in the diagram above.

(e)

2



Find the value of x giving reasons.

Examination continues overleaf ...

QUESTION ELEVEN (Continued)

(f) Find:

(i) $\int (x^2 - x + 2) dx$ 1

(ii) $\int \sin(3x + 1) dx$ 1

(g) Simplify $\frac{x^4 - x^2}{x - 1}$. 2

(h) Differentiate $y = \frac{2x}{1 + x^2}$. 2

QUESTION TWELVE (15 marks) Use a separate writing booklet.

Marks

- (a) Sketch the following functions, showing any intercepts or asymptotes. Sketch on separate axes.

(i) $y = -2x + 3$

2

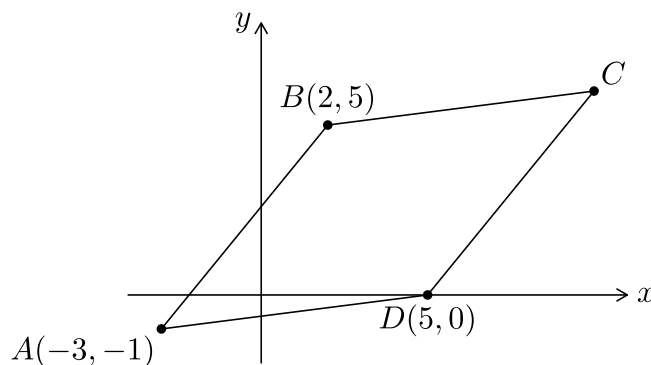
(ii) $y = \frac{1}{x - 4}$

2

(iii) $y = 2^{-x} + 2$

2

- (b)



In the diagram above, $ABCD$ is a parallelogram.

- (i) Find the co-ordinates of C if C lies in the first quadrant.
 (ii) Find the equation of line DC .
 (iii) Find the distance from B to line DC .
 (iv) Find the area of parallelogram $ABCD$.
 (v) Is $ABCD$ a rhombus? Give a reason for your answer.

2

2

2

2

1

QUESTION THIRTEEN (15 marks) Use a separate writing booklet.**Marks**

- (a) Solve the equation $5^{1-x} = 25^x$. **1**
- (b) A geometric series has common ratio of $-\frac{2}{3}$ and a limiting sum of 12. **2**
Find the first term.
- (c) Let R be the region bounded by the line $y = x$ and the parabola $y = 4x - x^2$
- (i) Sketch and shade the region R and find the x co-ordinates of the points of intersection of the line and the parabola. **3**
- (ii) Find the area of the region R . **2**
- (d) Two men leave point P . One travels 12 km on a bearing of 40°T and stops at point A . The second man travels 15 km on a bearing of 280°T and stops at point B .
- (i) Represent this situation on a diagram. **1**
- (ii) Find the size of angle APB . **1**
- (iii) Find the distance AB correct to the nearest 100 metres. **2**
- (iv) Find the true bearing of A from B to the nearest degree. **3**

QUESTION FOURTEEN (15 marks) Use a separate writing booklet.**Marks**

- (a) Solve the equation $2x^2 - 2x = 0$. **1**
- (b) Solve $|2x - 3| = x$. **2**
- (c) (i) Sketch $y = \tan x$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$. **1**
- (ii) Find the equation of the normal to the curve $y = \tan x$ at the point where $x = \frac{\pi}{4}$. **2**
- (d) The quadratic equation $2x^2 - 3px + 36 = 0$ has roots α and 2α , where $\alpha > 0$.
- (i) Find the value of α . **1**
- (ii) Find the value of p . **1**
- (e) A parabola has equation $y^2 - 4y = 8x + 4$.
- (i) Find the focal length and the vertex. **2**
- (ii) Sketch the parabola showing the focus and directrix. **2**
- (f) Solve the equation $\cos \theta - 2 \cos \theta \sin \theta = 0$ for $0 \leq \theta \leq 2\pi$. **3**

Examination continues next page ...

QUESTION FIFTEEN (15 marks) Use a separate writing booklet.

Marks

- (a) Use Simpson's rule with 3 function values to approximate $\int_1^3 2^{x^2} dx$. **2**
- (b) A particle is moving in a horizontal line and its displacement x metres from a fixed point O is given by $x = \frac{t^3}{3} - \frac{t^2}{2} - 6t + 11$ where t is measured in seconds. **3**
Find the average speed over the first 4 seconds.
- (c) A solid of revolution is formed when the region bounded by $y = \ln x$, the x -axis and $x = e^2$ is rotated about the y -axis.
- (i) Show the region to be rotated on a set of axes. **2**
- (ii) Find the volume of the solid of revolution in exact form. **3**
- (d) A university student plans to finish university in two years time. He wishes to travel overseas and thinks he will need \$15 000. He decides to invest \$ M at the start of each month in an investment account earning 3% per annum compounded monthly. Let A_n be amount in the account at the end of n months.
- (i) Find A_2 in terms of M . **1**
- (ii) Show that $A_n = 401M(1.0025^n - 1)$. **2**
- (iii) Find the amount he must invest each month to reach his target of \$15 000. Give your answer to the nearest dollar. **2**

QUESTION SIXTEEN (15 marks) Use a separate writing booklet.

Marks

- (a) The population P of a town was 12 500 at the beginning of the year 2000. Its population is growing at a rate proportional to its size, that is $\frac{dP}{dt} = kP$, where t is the time in years after 2000. At the beginning of 2015 its population was 16 000.

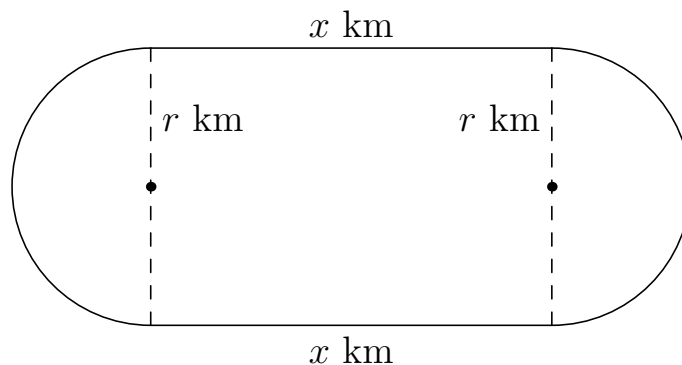
(i) Find the value of k .

2

(ii) What will the population be at the beginning of 2021?

2

(b)



A racetrack of perimeter one kilometre is formed from two equal semi-circles and a rectangle as shown in the diagram above. The straight sides have length x km and the circles have radius r km.

(i) Show that $x = \frac{1 - 2\pi r}{2}$.

1

(ii) Show that the area of the rectangle is $A = r - 2\pi r^2$.

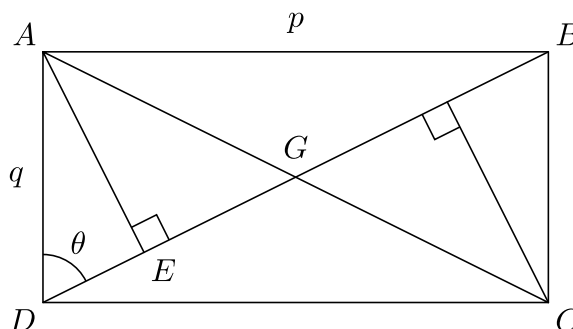
1

(iii) Find the values of x and r for which the area of the rectangle is maximised.

3

QUESTION SIXTEEN (Continued)

(c) The diagram below shows a rectangle $ABCD$ with $q < p$. Let $\theta = \angle ADE$.



(i) Show that $\cos \angle ADE = \frac{q}{\sqrt{p^2 + q^2}}$.

1

(ii) Show that $AE = \frac{pq}{\sqrt{p^2 + q^2}}$.

2

(iii) Show that the area of triangle AEG is $\frac{pq(p^2 - q^2)}{4(p^2 + q^2)}$.

3

————— End of Section II —————

END OF EXAMINATION

SolutionSection A (10 marks)

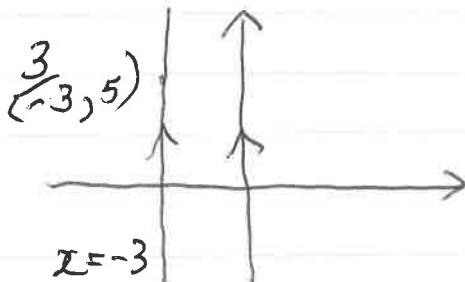
$$1 \quad \log_e 5 + \log_e 50$$

8

$$= 5.52 \quad (A)$$

$$2 \quad y = e^{2x+1}$$

$$y' = 2e^{2x+1} \quad (C)$$



(A)

$$6x\left(\frac{1}{x} + 1\right) = 0$$

$$x = 0 \quad \text{or} \quad \frac{1}{x} = -1$$

Not Valid

Isolation

(B)

$$4 \quad 2x+1 > 0$$

$$x > -\frac{1}{2}$$

(A)

$$5 \quad \frac{1}{2\sqrt{5}+3} \times \frac{2\sqrt{5}-3}{2\sqrt{5}-3}$$

$$= \frac{2\sqrt{5}-3}{20-9}$$

$$= \frac{2\sqrt{5}-3}{11}$$

(C)

$$9 \quad -x^2 - 6x - 5$$

$$= -(x^2 + 6x + 5)$$

$$= -(x^2 + 6x + 9) + 4$$

$$= -(x+3)^2 + 4$$

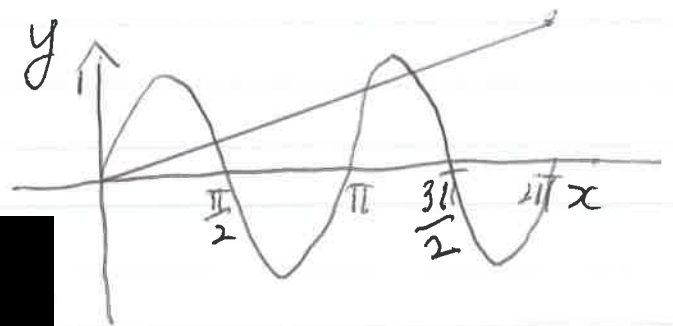
$$= 4 - (x+3)^2$$

(C)

$$6 \quad 3^x \times 2^x + 3x^{-2}$$

$$= 6^x + \frac{3}{x^2}$$

(D)



$$7 \quad 4 + e^{5x} = u,$$

$$u^2 - 11u + 10 = 0$$

$$(u-10)(u-1) = 0$$

$$e^{5x} = 10 \quad e^{5x} = 1$$

$$x = \frac{1}{5} \log_e 10 \quad x = 0$$

(D)

$$\sin 2x = \frac{1}{5} x$$

$$x = 2\pi \quad \frac{1}{5} 2\pi = 1.25$$

4 solutions

(C)

ELEVEN (15 marks)

②

15

$$(a) \quad \begin{aligned} 2 - 3x &\leq 11 \\ -3x &\leq 9 \\ x &\geq -3 \quad \checkmark \end{aligned}$$

$$(d) \quad \begin{aligned} \frac{x}{5} &= \frac{6}{4} \\ x &= \frac{30}{4} \\ x &= 7\frac{1}{2} \quad \checkmark \end{aligned}$$

$$(b) \quad \begin{aligned} 12, 6, 0, \dots \\ a &= 12, d = -6 \quad \checkmark \\ T_n &= a + (n-1)d \\ T_{41} &= 12 + 40 \times -6 \\ &= -228 \quad \checkmark \end{aligned}$$

$$(e) \quad \begin{aligned} 2x + 80 &= 120 \\ (\text{Ex + angle Th}) \quad \checkmark \\ 2x &= 40 \\ x &= 20 \quad \checkmark \end{aligned}$$

$$(f) \quad (i) \quad \int x^2 - x + 2 \, dx \\ = \frac{x^3}{3} - \frac{x^2}{2} + 2x + C \quad \checkmark$$

$$(c) \quad (i) \quad \begin{aligned} y &= 3x - x^{-1} \\ y' &= 3 + \frac{1}{x^2} \quad \checkmark \end{aligned}$$

$$(ii) \quad \int \sin(3x+1) \, dx \\ = -\frac{1}{3} \cos(3x+1) + C$$

$$(ii) \quad \begin{aligned} y &= \tan(2x+1) \\ y' &= \sec^2(2x+1) \times 2 \\ &= 2 \sec^2(2x+1) \quad \checkmark \end{aligned}$$

$$(g) \quad \begin{aligned} \frac{x^4 - x^2}{x-1} \\ &= \frac{x^2(x^2-1)}{x-1} \\ &= \frac{x^2(x-1)(x+1)}{x-1} \quad \checkmark \\ &= x^2(x+1) \quad \checkmark \end{aligned}$$

$$(iii) \quad \begin{aligned} y &= \ln(x^2+1) \\ y' &= \frac{2x}{x^2+1} \quad \checkmark \end{aligned}$$

$$(h) \quad \begin{aligned} y &= \frac{2x}{1+x^2} \\ y' &= \frac{vu' - uv'}{v^2} \\ &= \frac{(1+x^2)2 - 2x \times 2x}{(1+x^2)^2} \quad \checkmark \end{aligned}$$

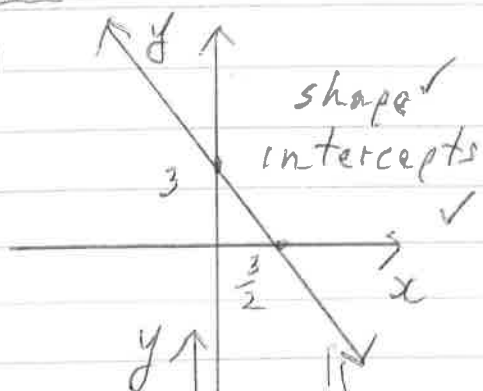
$$= \frac{2(1-x^2)}{(1+x^2)^2}$$

give marks here

(3)

TWELVE

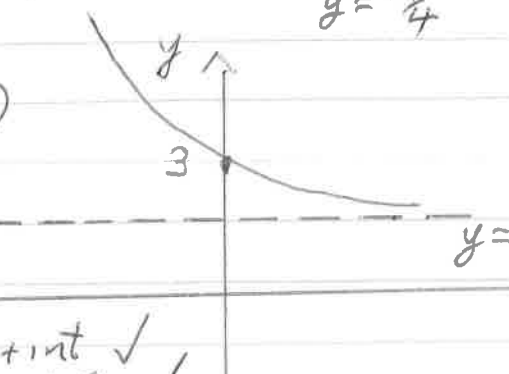
(a) (i)



(ii)

✓ shape
+ int

✓ asymptote

When $x=0$
 $y=-\frac{1}{4}$ 

(iii)

shape + int ✓
asymptote ✓(iii) Dist from B to DC
is $\frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}}$

(2, 5)

(x₁, y₁)

A = 6

B = -5

C = -30

$$= \frac{|6 \times 2 - 5 \times 5 - 30|}{\sqrt{6^2 + 5^2}}$$

$$= \frac{43}{\sqrt{61}}$$

(iv) Area of parallelogram

$$= b \times h$$

$$= |DC| \times \frac{43}{\sqrt{61}}$$

$$= \sqrt{61} \times \frac{43}{\sqrt{61}}$$

$$= 43 \text{ units}^2$$

$$(v) |AD| = \sqrt{8^2 + 1^2} \neq \sqrt{61}$$

Adjacent sides are
not equal ✓(b) (i) A → D across top
B → C same

$$\text{So } C = (10, 6) \checkmark$$

(ii)

$$m(DC) = \frac{\text{rise}}{\text{run}} = \frac{6}{5} \checkmark$$

Eqn of DC is

$$y - y_1 = m(x - x_1)$$

$$y - 0 = \frac{6}{5}(x - 5) \checkmark$$

$$y = \frac{6}{5}x - 6$$

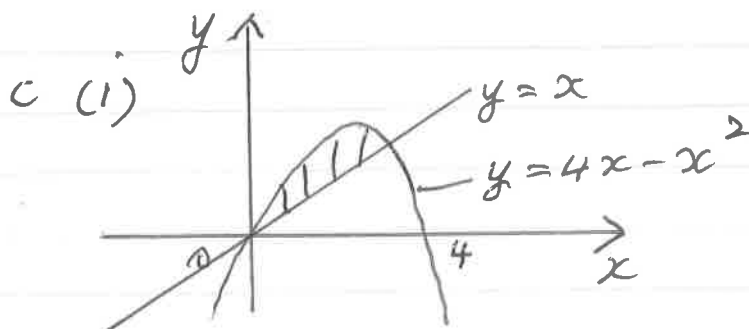
$$6x - 5y - 30 = 0$$

15

THIRTEEN

(a) $5^{1-x} = 25^x$
 $1-x = 2x$
 $x = \frac{1}{3}$ ✓

(b) $S_{\infty} = \frac{a}{1-r}$
 $12 = \frac{a}{1+\frac{2}{3}}$ ✓
 $12 = \frac{a}{\frac{5}{3}}$
 $a = 20$ ✓

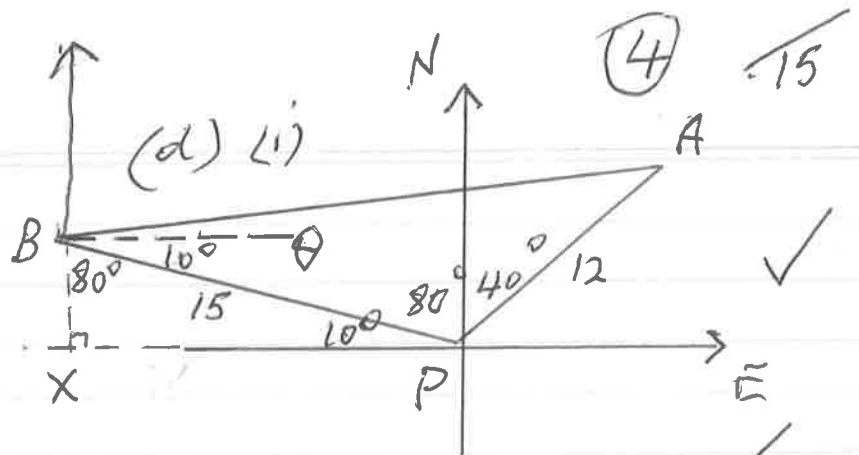


✓ correct graphs
 ✓ correct shading

(ii) Points of intersection
 where $x = 4x - x^2$

$x^2 - 3x = 0$
 $x(x-3) = 0$ ✓
 $x = 0, 3$

(iii) Area = $\int_0^3 (4x - x^2 - x) dx$
 $= \int_0^3 (3x - x^2) dx$ ✓
 $= \left[\frac{3x^2}{2} - \frac{x^3}{3} \right]_0^3$
 $= \frac{27}{2} - 9$
 $= \frac{9}{2}$ units² ✓



(ii) $\angle APB = 120^\circ$ ✓

(iv) $AB^2 = 15^2 + 12^2 - 2 \times 15 \times 12 \cos 120^\circ$ ✓
 $= 225 + 144 - 2 \times 180 \times -\frac{1}{2}$
 $= 369 + 180$
 $= 549$ ✓
 $AB = \sqrt{549} \approx 23.4 \text{ km}$

(iv) See diagram
 Let $\theta = \angle ABP$

$\cos \theta = \frac{AB^2 + BP^2 - AP^2}{2 \times AB \times BP}$ ✓

$\cos \theta = \frac{549 + 225 - 12^2}{2 \times 15 \times \sqrt{549}}$
 $\theta \approx 26.3^\circ$

Bearing of A from B ✓

$\approx 180^\circ - 80^\circ - 26.3^\circ$

(Note $\angle XBP = 80^\circ$)

$= 74^\circ \text{ T (to } \checkmark$

the nearest degree

✓ You can also use Sine rule ✓

FOURTEEN

15

(5)

(a) $2x^2 - 2x = 0$
 $2x(x-2) = 0$
 $x = 0$ or $x = 2$ ✓

(d) $2x^2 - 3px + 36 = 0$
 Roots are $\alpha, 2\alpha$
 (i) Product of roots
 $= 2\alpha^2 = \frac{36}{2}$
 $\alpha^2 = 9$ ✓
 $\alpha = 3$ ($\alpha > 0$)

(b) $|2x-3| = x$

$2x-3 = x$ or $2x-3 = -x$ (ii) Subst $\alpha = 3$

$x = 3$ or $x = 1$

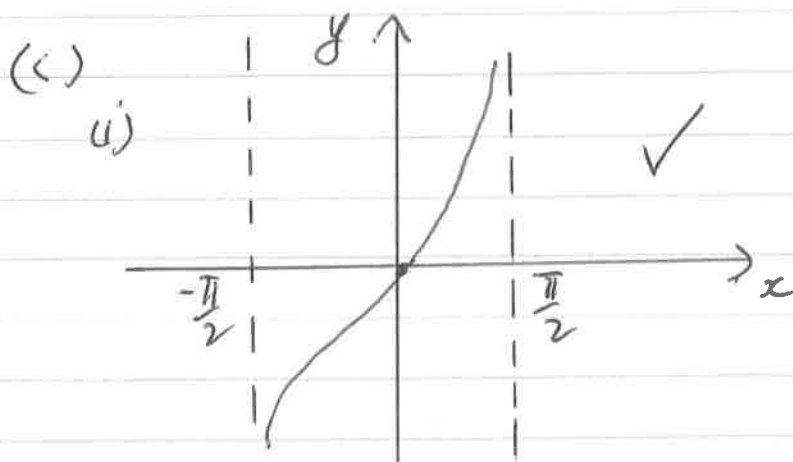
into the equation

$18 - 9p + 36 = 0$

$54 = 9p$ ✓

$p = 6$ ✓

[or use Σ roots]



(f) $\cos \theta - 2 \cos \theta \sin \theta = 0$
 $\cos \theta (1 - 2 \sin \theta) = 0$ ✓

$\cos \theta = 0$ or $\sin \theta = \frac{1}{2}$

(ii) $y = \tan x$

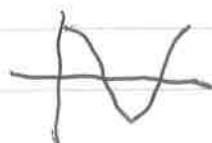
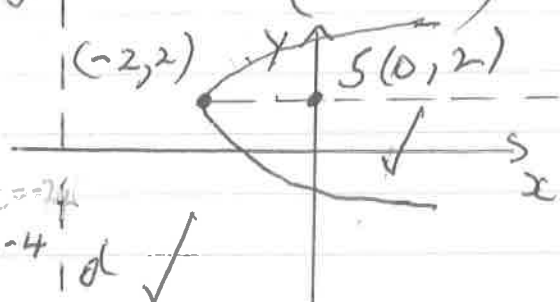
$y' = \sec^2 x$ ✓

When $x = \frac{\pi}{4}$, $M = \sec^2 \frac{\pi}{4} = 2$

$M_{\perp} = -\frac{1}{2}$, $y = 1$

Eqn of the normal is

$y - 1 = -\frac{1}{2} \left(x - \frac{\pi}{4} \right)$ ✓



$\theta = \frac{\pi}{2}, \frac{3\pi}{2}$ $\theta = \frac{\pi}{6}, \frac{5\pi}{6}$

✓

✓

(e) $y^2 - 4y = 8x + 4$

$y^2 - 4y + 4 = 8x + 8$

$(y-2)^2 = 4(2)(x+2)$ ✓

focal length = 2

Vertex = $(-2, 2)$ ✓

5(6)

FIFTEEN

15

$$(a) A = \frac{b-a}{6} \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right]$$

$$\therefore \int_1^3 2x^2 dx \doteq \frac{3-1}{6} \left[f(1) + 4f(2) + f(3) \right]$$

$$\left. \begin{array}{l} f(1) = 2 \\ f(2) = 16 \\ f(3) = 512 \end{array} \right\} \checkmark \quad \doteq \frac{1}{3} (2 + 4 \times 16 + 512)$$

$$\doteq \frac{578}{3} \doteq 192.6 \text{ m}^2 \quad \checkmark$$

[Award 1 wrong values entered into correct formula]

$$(b) x = \frac{t^3}{3} - \frac{t^2}{2} - 6t + 11$$

$$\frac{dx}{dt} = t^2 - t - 6$$

Velocity zero when

$$t^2 - t - 6 = 0$$

$$(t-3)(t+2) = 0$$

$$t = 3 \text{ or } t = -2 \quad \checkmark$$

When $t = 0$, $x = 11$

$$t = 3, \quad x = 9 - \frac{9}{2} - 18 + 11$$

$$= 20 - 22\frac{1}{2}$$

$$= -2\frac{1}{2} \quad \checkmark$$

$$t = 4, \quad x = \frac{64}{3} - 8 - 24 + 11$$

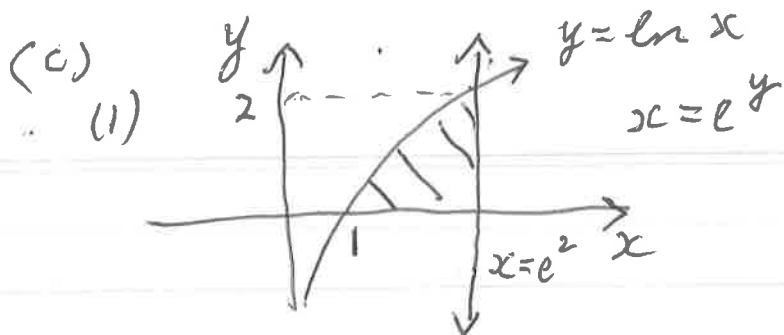
$$= \frac{1}{3}$$

In first 4 secs average speed = distance travelled

$$= \frac{13\frac{1}{2} + 2\frac{1}{2} + \frac{1}{3}}{4} \quad \checkmark$$

(= 4.083)
Accept approx

$$= 4\frac{1}{12} \text{ m/sec}$$



✓ graph of $y = \ln x$
 ✓ correct region

$$\begin{aligned}
 \text{(ii)} \quad V &= \pi \int_0^2 x_R^2 - x_L^2 dy \\
 &= \pi \int_0^2 e^{4y} - e^{2y} dy \quad \checkmark \\
 &= \pi \left[\frac{e^{4y}}{4} - \frac{e^{2y}}{2} \right]_0^2 \quad \checkmark \\
 &= \pi \left(\left(\frac{e^4}{4} - \frac{e^2}{2} \right) - \left(\frac{1}{4} - \frac{1}{2} \right) \right) \\
 &= \pi \left(\frac{e^4}{4} + \frac{1}{4} \right) \text{ units}^3 \quad \checkmark
 \end{aligned}$$

(d) (i) $A_1 = 1.0025M$ ✓

$A_2 = 1.0025M + 1.0025^2M$

(ii) $A_3 = 1.0025M + 1.0025^2M + 1.0025^3M$

$A_n = (1.0025 + 1.0025^2 + \dots + 1.0025^n)M$

$= M \times \frac{r^n - 1}{r - 1}$ ✓

$= M \times \frac{1.0025^n - 1}{0.0025}$

✓ $= 401M (1.0025^n - 1)$ ✓

$A_{24} = 401M (1.0025^{24} - 1) = 15000$

$M = \$606$ (to nearest dollar) ✓

(7)

SIXTEEN

$$(a) \quad \frac{dP}{dt} = kP$$

$$(i) \quad P = P_0 e^{kt}$$

$$P_0 = 12500$$

$$t = 15 \quad P = 16000$$

$$16000 = 12500 e^{k(15)}$$

$$\therefore k = \frac{1}{15} \ln \frac{32}{25} \quad \checkmark$$

$$(ii) \text{ When } t = 20$$

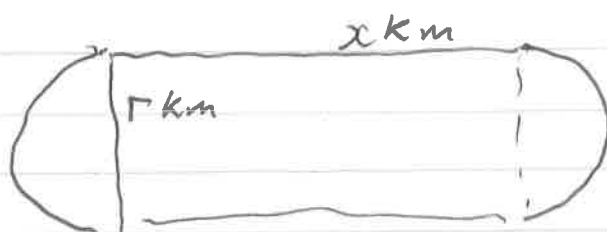
$$P = 12,500 e^{\frac{4}{3} \log_e \left(\frac{32}{25} \right)}$$

$$= 12,500 \times \left(\frac{32}{25} \right)^{\frac{4}{3}}$$

$$\approx 17370 \text{ (nearest 10)} \quad \checkmark$$

(Accept correct to the nearest hundred)

(b)



(i)

$$\text{Perimeter} = 2\pi r + 2x = 1$$

$$x = \frac{1 - 2\pi r}{2} \quad \checkmark$$

(ii)

$$A = \cdot 2r x$$

$$(\text{Rectangle}) = 2r \left(\frac{1 - 2\pi r}{2} \right) \quad \checkmark$$

$$= r - 2\pi r^2$$

(upside parabola so has max)

(iii)

$$A' = 1 - 4\pi r$$

$$A'' = -4\pi < 0 \quad \checkmark$$

$$\text{Max where } A' = 0$$

$$1 - 4\pi r = 0$$

$$r = \frac{1}{4\pi}$$

$$x = \frac{1}{4} \quad \checkmark$$

SIX TEEN

(a) $\frac{dP}{dt} = kP$
 (i) $P = P_0 e^{kt}$

15

$P_0 = 12500$
 $t = 15$ $P = 16000$ ✓
 $16000 = 12500 e^{k(15)}$
 $\therefore k = \frac{1}{15} \ln \frac{32}{25}$ ✓

(ii) When $t = 21$
 $P = 12,500 e^{\frac{21}{15} \ln \left(\frac{32}{25}\right)}$ ✓
 $= 12,500 \times \left(\frac{32}{25}\right)^{\frac{21}{15}}$ ✓
 ≈ 17660 (nearest 10) ✓
 (Accept correct to the nearest hundred)

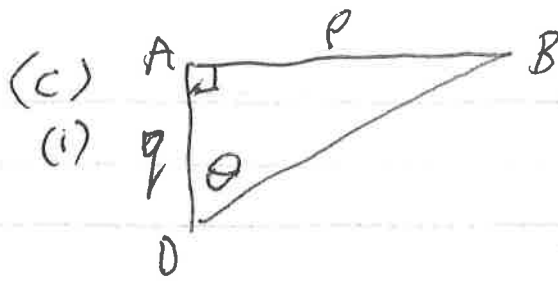


(i) Perimeter $= 2\pi r + 2x = 1$
 $x = \frac{1 - 2\pi r}{2}$ ✓

(ii) $A = 2rx$
 (Rectangle) $= 2r \left(\frac{1 - 2\pi r}{2} \right)$ ✓

$= r - 2\pi r^2$ (upside parabola so has max)
 (iii) $A' = 1 - 4\pi r$
 $A'' = -4\pi < 0$ ✓
 Max where $A' = 0$
 $1 - 4\pi r = 0$ ✓
 $r = \frac{1}{4\pi}$ ✓
 $x = \frac{1}{4}$ ✓

(8)



By Pythagoras'

$$BD = \sqrt{p^2 + q^2}$$

$$\cos \theta = \frac{p}{\sqrt{p^2 + q^2}} \quad \checkmark$$

(ii)

$$\text{Area } \triangle ADB = \frac{1}{2} p q \quad \checkmark$$

$$\text{Also Area } \triangle ADB = \frac{1}{2} AE \times DB = \frac{1}{2} p q$$

$$AE = \frac{p q}{DB} = \frac{p q}{\sqrt{p^2 + q^2}} \quad \checkmark$$

(There are other ways to do this)

(iii)

$$AG = \frac{1}{2} AC = \frac{1}{2} \sqrt{p^2 + q^2} \quad (\text{Pythagoras})$$

$$\begin{aligned} EG^2 &= AG^2 - AE^2 \\ &= \frac{1}{4} (p^2 + q^2) - \frac{p^2 q^2}{p^2 + q^2} \quad \checkmark \\ &= \frac{p^4 + 2p^2 q^2 + q^4 - 4p^2 q^2}{4(p^2 + q^2)} \end{aligned}$$

$$= \frac{p^4 - 2p^2 q^2 + q^4}{4(p^2 + q^2)}$$

$$EG^2 = \frac{p^2 - q^2}{4(p^2 + q^2)} \quad \checkmark$$

$$EG = \frac{|p^2 - q^2|}{2\sqrt{p^2 + q^2}} = \frac{p^2 - q^2}{2\sqrt{p^2 + q^2}} \quad \text{since } p > q$$

$$\begin{aligned} \text{Area } \triangle AEG &= \frac{1}{2} AE \times EG \\ &= \frac{1}{2} \frac{p q}{\sqrt{p^2 + q^2}} \times \frac{(p^2 - q^2)}{2\sqrt{p^2 + q^2}} \quad \checkmark \\ &= \frac{p q (p^2 - q^2)}{4(p^2 + q^2)} \end{aligned}$$